

Endpoint Control Does Not Tell You Which Earlier Step Still Matters

Human-readable companion to Endpoint Control Does Not Identify Temporal Causal Factorization

Human-readable companion to the completed scientific manuscript. This version preserves the scientific claims and limits of the completed manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.

Abstract

A later outcome can show that an earlier sequence had causal power without telling us which time-located parts of that sequence still make a causal difference at the endpoint. This paper formalizes that gap. A complete intervention history can be compressed into one compound input to an endpoint channel, and an aggregate quantity such as channel capacity can correctly describe how much distinguishable influence those complete histories have over the endpoint. But that abstraction can forget how the histories were factored into earlier and later interventions. The paper proves that endpoint-equivalent systems can differ in whether a fixed earlier coordinate remains consequential at the endpoint. A deterministic witness, a stochastic extension, and exhaustive finite enumeration show that the ambiguity is structural rather than a quirk of one scalar summary. The result is methodological, not a new cognitive mechanism: matched endpoint control does not identify matched temporal causal credit. When a cognitive theory asks which earlier action, cue, or perturbation still matters, it must retain or independently recover the temporal factorization.

1. The human problem: the outcome changed, but what deserves credit?

Suppose an experiment shows that changing a sequence of actions changes a later outcome. That is real causal evidence. But now ask a more specific question: which earlier action in the sequence is still causally relevant to the endpoint? The answer does not automatically come with the endpoint effect.

This is the intuitive puzzle behind ButWhy. The same final amount of control can be produced by histories whose earlier parts are organized differently. One early intervention may remain consequential all the way to the endpoint in one system, while a corresponding early difference in another system may matter temporarily and then reconverge before the endpoint.

Credit-assignment research already asks how earlier actions or states should be associated with later outcomes, and temporally resolved causal or information-theoretic tools already exist (Moran et al., 2021; Pignatelli et al., 2024; Ay & Polani, 2008). The paper does not add a new credit-assignment algorithm. It asks a prior identification question: what can an endpoint-level control description establish before any particular cognitive credit mechanism is assumed?

2. The compression that creates the ambiguity

Start with a time-factored intervention history: an action or intervention at time 0, another at time 1, and so on. The temporal coordinates are part of the scientific description because “the early action” and “the late action” can have different theoretical meanings.

$$U = (A_0, A_1, \dots, A_{T-1})$$

Now treat the entire history U as one atomic input symbol to a channel whose output is the later endpoint ET . Nothing has changed dynamically. The same full causal mapping is being written in a coarser way. But the representation no longer has to preserve which part of the compound symbol came from which temporal coordinate.

Endpoint channel capacity then asks how much distinguishable influence the compound histories can exert on the endpoint. This is a legitimate control quantity. In a deterministic system it depends on how many endpoint classes can be reached. It does not, by itself, tell us how the time-factored histories are arranged inside those endpoint classes.

3. The stronger question: does a fixed earlier coordinate still make a difference?

To ask whether an earlier intervention still matters, the manuscript holds all other time coordinates fixed and changes only the coordinate of interest. If that controlled change alters the endpoint, the coordinate has endpoint causal support in that context. If both histories end at the same endpoint, the earlier difference has reconverged with respect to that endpoint.

This is not the same as marginal mutual information. A coordinate can have zero marginal mutual information with the endpoint yet still change the endpoint under controlled contrasts because of interaction or synergy. The paper therefore uses a controlled intervention-support relation rather than treating simple marginal association as causal relevance.

The core identification question is whether a summary that is indifferent to how complete histories are renamed can recover a target whose meaning depends on preserving the temporal coordinates.

4. Two kinds of “same system” are doing different jobs

At the abstract endpoint-channel level, complete histories are just input symbols. Any one-to-one relabeling of those complete symbols leaves channel capacity unchanged. If two channels differ only by such a row permutation, the endpoint abstraction regards them as equivalent.

A time-sensitive causal question permits a narrower relabeling. It is harmless to rename the values available within a fixed time coordinate—for example, swapping the labels assigned to the two actions at time 0. It is not harmless to arbitrarily exchange or mix complete histories in ways that destroy which distinction belonged to which temporal coordinate.

Equivalence	What may be relabeled	What it preserves
Abstract compound-input equivalence	Whole intervention histories can be permuted as atomic symbols	Endpoint-channel quantities such as capacity
Time-preserving equivalence	Values may be relabeled within fixed temporal coordinates	Structural questions about which time-indexed intervention distinctions remain consequential

The endpoint abstraction therefore has a larger symmetry than the temporal question. That symmetry mismatch is where the information can be lost.

5. The identification result

The mathematical logic is standard invariant/orbit reasoning rather than a new theorem of invariance. If a representation is constant under a class of transformations, but the scientific target changes within that same equivalence class, the representation cannot recover that target in general (Chiu et al., 2025).

Applied here: endpoint-control statistics such as channel capacity are invariant when whole compound histories are arbitrarily relabeled. The temporal-support target is only invariant under the narrower transformations that preserve the time factorization. If two endpoint-equivalent channels fall into different time-preserving support classes, endpoint control cannot identify which temporal support structure is present.

Endpoint causal capacity does not identify temporal causal factorization.

The result is stronger than the generic statement that a scalar loses information. Even retaining the endpoint channel up to abstract input-symbol identity can be insufficient. The missing datum is the embedding of those atomic history symbols into the temporal product structure.

6. A concrete witness: same endpoint channel, different early causal support

The manuscript constructs two deterministic three-step systems. As abstract endpoint channels, one is obtained from the other by permuting the rows associated with complete intervention histories. They therefore have the same deterministic endpoint capacity and the same aggregate endpoint information under the stated probing distribution.

But their first temporal coordinates differ maximally. In one system, changing A0 while holding the later coordinates fixed changes the endpoint in every relevant context: first-coordinate support is full. In the other, changing A0 never changes the final endpoint: the earlier distinction appears in the trajectory but has completely reconverged by the endpoint.

No relabeling confined to values within fixed time coordinates can turn a full support set into an empty one. The two systems are therefore identical under the coarser abstract endpoint-channel equivalence while remaining different under the factor-preserving temporal equivalence.

Passing both systems through the same informative stochastic post-channel does not remove the distinction. Their aggregate endpoint channels remain equivalent under the same input relabeling, while the controlled early-coordinate contrast remains detectable in one and absent in the other until the output channel is degraded to the point where all endpoint distinctions disappear.

7. The ambiguity is not rare in the small finite case

The paper then exhaustively enumerates balanced deterministic mappings from three binary intervention coordinates to four equally occupied endpoint classes. Every mapping in this family has the same two-bit deterministic endpoint capacity, yet the earlier coordinates can have very different structural relationships to the endpoint.

After discarding arbitrary endpoint names, 105 unlabeled balanced partitions remain. Under the time-preserving relabelings that may rename action values inside each fixed coordinate, those partitions still form 35 distinct structural equivalence classes. Even allowing permutations of coordinate positions—a stronger equivalence that deliberately discards temporal identity—still leaves multiple classes.

The paper further shows that adding a simple sensitivity count for each temporal coordinate is not enough to reconstruct the full organization. Different endpoint partitions can share the same coordinate-sensitivity profile. So temporal causal structure is not exhausted by assigning one influence magnitude to each time point.

8. What existing science already owns

The components around the result are heavily precedented. Empowerment is an established information-theoretic measure of potential control (Klyubin et al., 2005, 2008), and richer process-empowerment formulations explicitly preserve more temporal structure (Tiomkin et al., 2025). Temporal credit assignment is a mature computational problem, and human work shows sensitivity to causal and task structure in assigning credit (Moran et al., 2021; Pignatelli et al., 2024).

Causal-abstraction work already emphasizes preserving intervention distinctions relevant to the intended query, and variable/product structure is already treated as causally significant (Beckers & Halpern, 2019; Rubenstein et al., 2017; Park et al., 2023; Buchholz et al., 2024; Xia & Bareinboim, 2025). Group invariance and maximal-invariant mathematics are also established. The manuscript therefore does not claim a new generic invariance theorem, a new causal-information measure, or a new general theory of control.

The residual contribution is narrow: the specific mismatch between abstract endpoint-history relabeling and time-preserving factorization; the explicit endpoint-isomorphic witness with full-versus-empty fixed-coordinate support; its stochastic persistence; the exhaustive finite ambiguity analysis; and the scaling family. Exact prior-art ownership of that integrated application remains OPEN.

9. What this means for cognitive experiments

If a cognitive experiment shows that a manipulation changes a delayed endpoint, the experiment has established an endpoint causal relation under its design. It has not thereby established which earlier action, cue, policy update, or perturbation remains causally relevant at that endpoint.

A stronger process-level claim requires retaining the temporal coordinates or using a richer causal representation capable of distinguishing persistence from reconvergence. This matters whenever a theory assigns different scientific roles to earlier and later events rather than treating the whole history as one indivisible input.

The manuscript proposes one downstream empirical question but does not answer it: if two task conditions are matched on reward, future opportunity, and aggregate endpoint control, does later choice or learning differ depending on whether the agent’s own earlier causal contribution persists to the endpoint or reconverges before it? That behavioral sensitivity hypothesis is OPEN and no participant evidence is supplied.

10. What the result does not establish

- It does not show that a cognitive system represents the temporal-support relation.
- It does not supply a psychological or neural credit-assignment mechanism, gate, selector, planning algorithm, temporal-depth estimator, or successor representation.
- It does not establish that consciousness detects or controls the distinction.
- It does not show that organisms or artificial agents prefer persistent causal contribution over reconvergence.
- It does not recover complete causal provenance: mediation, feedback, interaction, higher-order synergy, and cross-boundary pathways can require richer models.

The formal result would remain valid even if no organism were behaviorally sensitive to the distinction. Its immediate claim is methodological: a coarser endpoint abstraction and a time-factored causal query do not identify the same object.

11. When the distinction should disappear

The result is conditional on a scientifically meaningful temporal factorization. If the coordinates have no relevant separate meaning and complete histories genuinely should be treated as atomic symbols, then the additional temporal structure is irrelevant by construction. The theorem is not a demand to factor every process.

The mathematical application would also fail if the witness systems were not truly equivalent under the endpoint abstraction being invoked, or if their factor-sensitive support structures occupied the same time-preserving equivalence class. The manuscript verifies that neither failure occurs for its constructions.

For cognitive applications, the practical loss rule is simpler: if the theory asks only how much endpoint control the whole history possesses, endpoint control is the right object. If it asks which earlier intervention still matters, endpoint equivalence alone is insufficient and the temporal factorization must be preserved or independently recovered.

12. Conclusion

Endpoint control and temporal causal credit are related questions, but they are not the same question. Compressing a sequence into one compound history can preserve exactly how much distinguishable influence the history has over the endpoint while forgetting how that influence is distributed across the time-labeled coordinates.

The formal result shows that this is a genuine identification limit, not just vague underdetermination. Systems can be identical under the endpoint abstraction and different in whether a fixed earlier intervention remains consequential. The difference survives noise and persists across many small systems with matched endpoint capacity.

Accordingly, when a cognitive theory says an earlier action, cue, or perturbation still matters because the overall sequence controlled a later outcome, one more step is required. The experiment must preserve or recover the temporal factorization relevant to that claim. The endpoint cannot supply it by itself.

References

- Abel, D., Bowling, M., Barreto, A., Dabney, W., Dong, S., Hansen, S., Harutyunyan, A., Khetarpal, K., Lyle, C., Pascanu, R., Piliouras, G., Precup, D., Richens, J., Rowland, M., Schaul, T., & Singh, S. P. (2025). Plasticity as the Mirror of Empowerment. *Advances in Neural Information Processing Systems*, 38. <https://doi.org/10.52202/085713-5481>
- Ay, N., & Polani, D. (2008). Information flows in causal networks. *Advances in Complex Systems*, 11(1), 17-41.
- Barnett, N., & Crutchfield, J. P. (2015). Computational mechanics of input-output processes: Structured transformations and the epsilon-transducer. *Journal of Statistical Physics*, 161(2), 404-451. <https://doi.org/10.1007/s10955-015-1327-5>
- Beckers, S., & Halpern, J. Y. (2019). Abstracting causal models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 2678-2685. <https://doi.org/10.1609/aaai.v33i01.33012678>

- Buchholz, S., Park, J., & Schölkopf, B. (2024). Products, abstractions and inclusions of causal spaces. *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence, Proceedings of Machine Learning Research*, 244, 430-449. <https://proceedings.mlr.press/v244/buchholz24a.html>
- Chiu, K., Sharp, A., & Bloem-Reddy, B. (2025). Randomization tests for conditional group symmetry. *Electronic Journal of Statistics*, 19(2), 5979-6039. <https://doi.org/10.1214/25-EJS2473>
- Kahn, J., Kalai, G., & Linial, N. (1988). The influence of variables on Boolean functions. *Proceedings of the 29th Annual Symposium on Foundations of Computer Science*, 68-80. <https://doi.org/10.1109/SFCS.1988.21923>
- Klyubin, A. S., Polani, D., & Nehaniv, C. L. (2005). Empowerment: A universal agent-centric measure of control. *Proceedings of the IEEE Congress on Evolutionary Computation*, 128-135.
- Klyubin, A. S., Polani, D., & Nehaniv, C. L. (2008). Keep your options open: An information-based driving principle for sensorimotor systems. *PLoS ONE*, 3(12), e4018. <https://doi.org/10.1371/journal.pone.0004018>
- Moran, R., Dayan, P., & Dolan, R. J. (2021). Human subjects exploit a cognitive map for credit assignment. *Proceedings of the National Academy of Sciences of the United States of America*, 118(4), e2016884118. <https://doi.org/10.1073/pnas.2016884118>
- Park, J., Buchholz, S., Schölkopf, B., & Muandet, K. (2023). A measure-theoretic axiomatisation of causality. *Advances in Neural Information Processing Systems*, 36, 28510-28540. https://proceedings.neurips.cc/paper_files/paper/2023/hash/5aadf1e309cc03cab3ec35afb7c9d0c8-Abstract-Conference.html
- Pignatelli, E., Ferret, J., van Hasselt, H., Geist, M., Mesnard, T., Pietquin, O., & Toni, L. (2024). A survey of temporal credit assignment in deep reinforcement learning. *Transactions on Machine Learning Research*.
- Rubenstein, P. K., Weichwald, S., Bongers, S., Mooij, J. M., Janzing, D., Grosse-Wentrup, M., & Schölkopf, B. (2017). Causal consistency of structural equation models. *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence*.
- Tiomkin, S., Salge, C., & Polani, D. (2025). Process empowerment for robust intrinsic motivation. *Journal of Physics: Complexity*, 6, 035011.
- Xia, K. M., & Bareinboim, E. (2025). Causal abstraction inference under lossy representations. *Proceedings of the 42nd International Conference on Machine Learning, Proceedings of Machine Learning Research*, 267, 68225-68235.
- Yiu, E., Allen, K. R., Ginosar, S., & Gopnik, A. (2026). Empowerment gain and causal model construction: Children and adults are sensitive to controllability and variability in their causal interventions. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 384(2320), 20250003. <https://doi.org/10.1098/rsta.2025.0003>